



PGPDSE FT Capstone Project – Interim Report

On

Project Title:

Uncovering Success Factors of Local Businesses

Through Yelp Data Analytics

Batch details	PGP in Data Science with Specialization in GenAI July 2025 - February 2026
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TABLE OF CONTENTS:

Title	Page No.
1. Industry Review	1
1.1 Background & Current Industry Practices	1
1.2 Literature Survey	2
2. Dataset and Domain Understanding	3
2.1 Domain	3
2.2 Dataset Description	3
2.3 Data Dictionary (Key Variables)	4
2.4 Variable Categorization	4
2.5 Pre-Processing & Data Quality Analysis	5
2.6 Alternate Data Sources (Future Scope)	6
2.7 Project Justification	6
3. Data Exploration (EDA)	7
3.1 Relationship Between Variables	7
3.2 Multi-Collinearity	9
3.3 Pairplot for Key Numerical Variables	10
3.4 Distribution Analysis	11
3.5 Outliers	12
3.6 Statistical Significance	13
3.7 Class Imbalance	14
4. Feature Engineering	15
4.1 Transformations Applied	15
4.2 Scaling	16
4.3 Feature Selection	16
4.4 Dimensionality Reduction	17
4.5 Feature Leakage Prevention	17

Title	Page No.
4.6 Model Readiness Validation	18
5. Assumptions	19
5.1 Regression Models	19
5.2 Classification Models	20
5.3 Clustering	21

LIST OF FIGURES:

Figure No.	Figure Name	Page No.
Figure 1	Stars vs Review Count	7
Figure 2	Price Range vs Average Rating	7
Figure 3	Attributes Impact on Ratings	8
Figure 4	Correlation Matrix	9
Figure 5	Pairplot for Key Numerical Variables	10
Figure 6	Rating Distribution	11
Figure 7	Review Count Distribution	11
Figure 8	Outliers Detection for Review Count	12
Figure 9	Class Imbalance	14

LIST OF TABLES:

Table No.	Table Name	Page No.
Table 1	Data Dictionary	4
Table 2	Variable Categorization	4
Table 3	Future Data Analysis	6
Table 4	Correlation Table	9
Table 5	Class Imbalance	13
Table 6	Data Transformation	15

1. Industry Review

1.1 Background & Current Industry Practices

Online review platforms such as **Yelp** play a critical role in influencing customer decisions and business success. Businesses rely heavily on customer ratings, reviews, and service attributes to attract and retain customers. However, most platforms provide raw data without structured insights, making it difficult for business owners to understand **what truly drives customer satisfaction**.

In the current industry practice:

- Businesses use **average ratings** as a performance metric.
 - Decisions are often **intuition-based**, not data-driven.
 - Limited analytics are available to explain rating variations.
 - Advanced analytics and AI-driven insights are increasingly being adopted to understand consumer behaviour, improve service quality, and enhance customer engagement.
 - Online ratings directly influence a business's **visibility and ranking** on digital platforms.
 - Customer expectations vary across **price ranges and service types**, making generalized ratings insufficient.
 - Small and medium businesses often lack **analytical tools** to interpret customer feedback effectively.
 - Understanding attribute-level impact enables **targeted operational improvements** rather than reactive responses.
-

1.2 Literature Survey

Previous research highlights:

- Online ratings significantly impact revenue and survival of businesses.
- Service quality attributes (delivery, WiFi, seating, pricing) influence customer satisfaction.
- Machine Learning models can effectively predict consumer ratings from structured attributes.
- Studies show that **higher-rated businesses attract more customers and online engagement**.
- Research highlights the effectiveness of **regression and tree-based models** for rating prediction.
- There is limited emphasis on **interpretable, attribute-driven insights** in existing studies.

Past studies mostly focus on **review text mining**, whereas this project focuses on **structured business attributes combined with Gen AI insights**, making it unique. This project bridges this gap by leveraging structured business attributes and GenAI-driven interpretation to generate actionable insights for business decision-making.

2. Dataset and Domain Understanding

2.1 Domain

- **Domain:** Online Business Reviews & Analytics
 - **Industry:** Food, Retail, Services & Local Businesses
-

2.2 Dataset Description

- **Source:** Yelp Open Dataset
 - **Size:** ~114,000 businesses
 - **Type:** Structured tabular dataset
-

2.3 Data Dictionary (Key Variables)

Category	Variables
Business Info	Name, City, State, Is_Open
Ratings	Stars, Review_Count
Pricing	PriceRange
Attributes	Delivery, TakeOut, WiFi, OutdoorSeating, GoodForKids, Caters
Categories	Category_Bins, Primary_Category_Bin
Target	Stars

Table 1: Data Dictionary

2.4 Variable Categorization

Type	Count
Numerical	5
Categorical	4
Boolean	10
Derived	6

Table 2: Variable Categorization

2.5 Pre-Processing & Data Quality Analysis

- Raw JSON attributes were parsed to extract meaningful service-level features such as Delivery, TakeOut, WiFi, and OutdoorSeating.
- The total number of categories per business was derived as a numerical feature (No_of_Categories) to capture business diversity.
- Extracted and cleaned key attributes.
- Converted categorical attributes using one-hot encoding to ensure compatibility with machine learning models.
- Missing values identified mainly in optional attributes (expected for real-world data).
- Attribute extraction from semi-structured data.
- Encoding methods (binary & one-hot).
- Missing value handling logic.

Missing values pattern:

Optional business attributes show 10–70% missing; core fields have 0% missing.

2.6 Alternate Data Sources (Future Scope)

Supplementary Data	Purpose
Review Text Data	Sentiment & semantic analysis
Temporal Data (year/month)	Trend analysis
Competitor Density	Local competition effects

Table 3: Future Data Analysis

2.7 Project Justification

Problem Statement:

Businesses lack clarity on which attributes and operational factors influence customer ratings and satisfaction.

Complexity Involved:

- High dimensional data
- Missing attributes
- Mixed variable types
- Subjective nature of ratings
- Data Cleaning for Json Attributes

Project Outcome:

- **Commercial Value:** Helps businesses improve services strategically.
- **Academic Value:** Demonstrates ML, statistics & Gen AI integration.
- **Social Value:** Enhances customer experience indirectly.

3. Data Exploration (EDA)

3.1 Relationship Between Variables

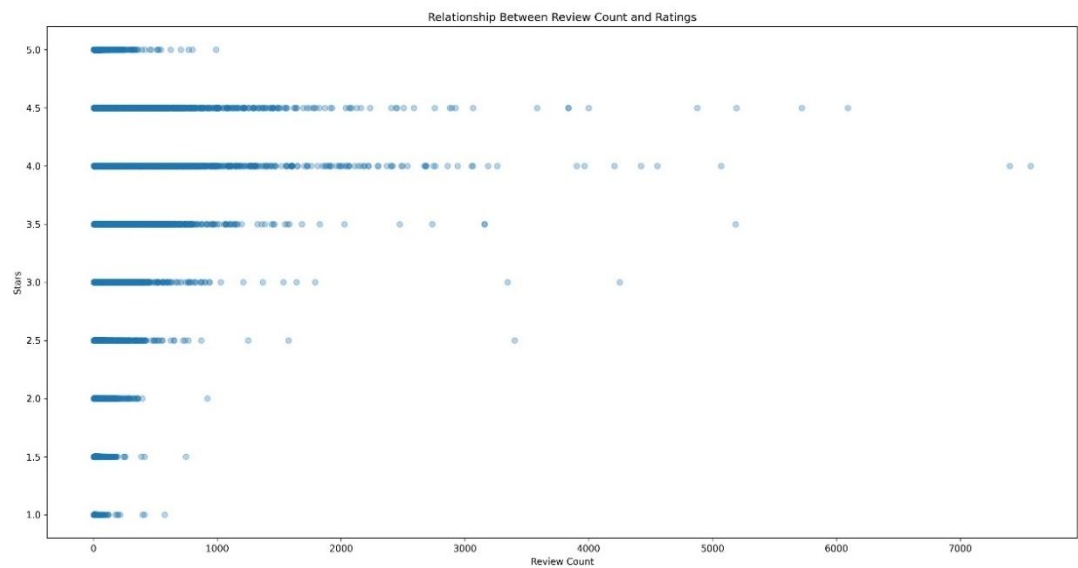


Figure 1: Stars vs Review Count

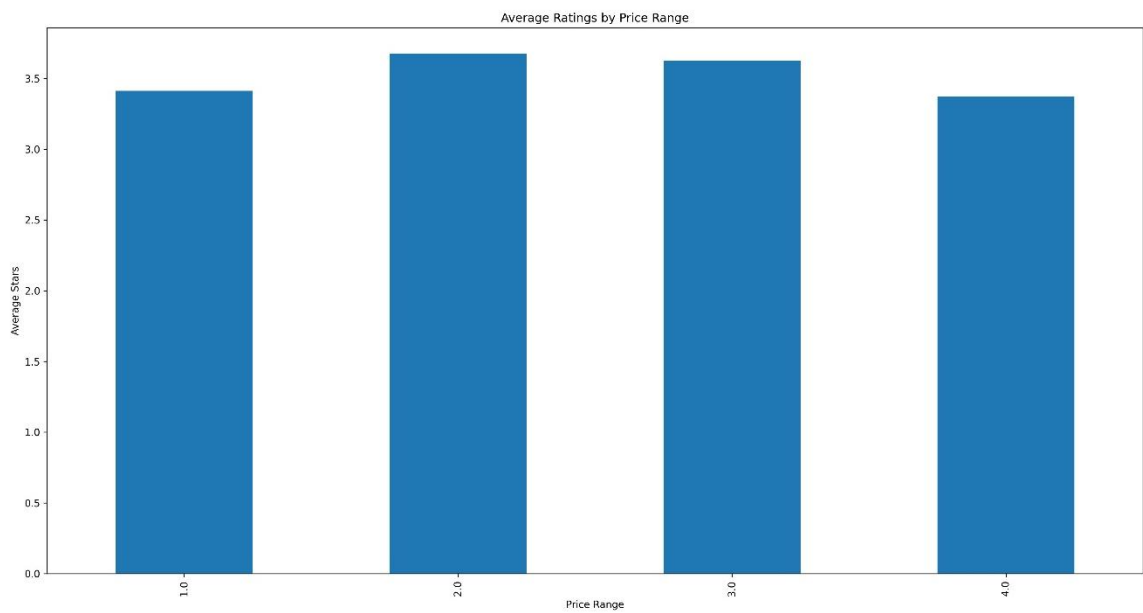


Figure 2: Price Range vs Average Rating

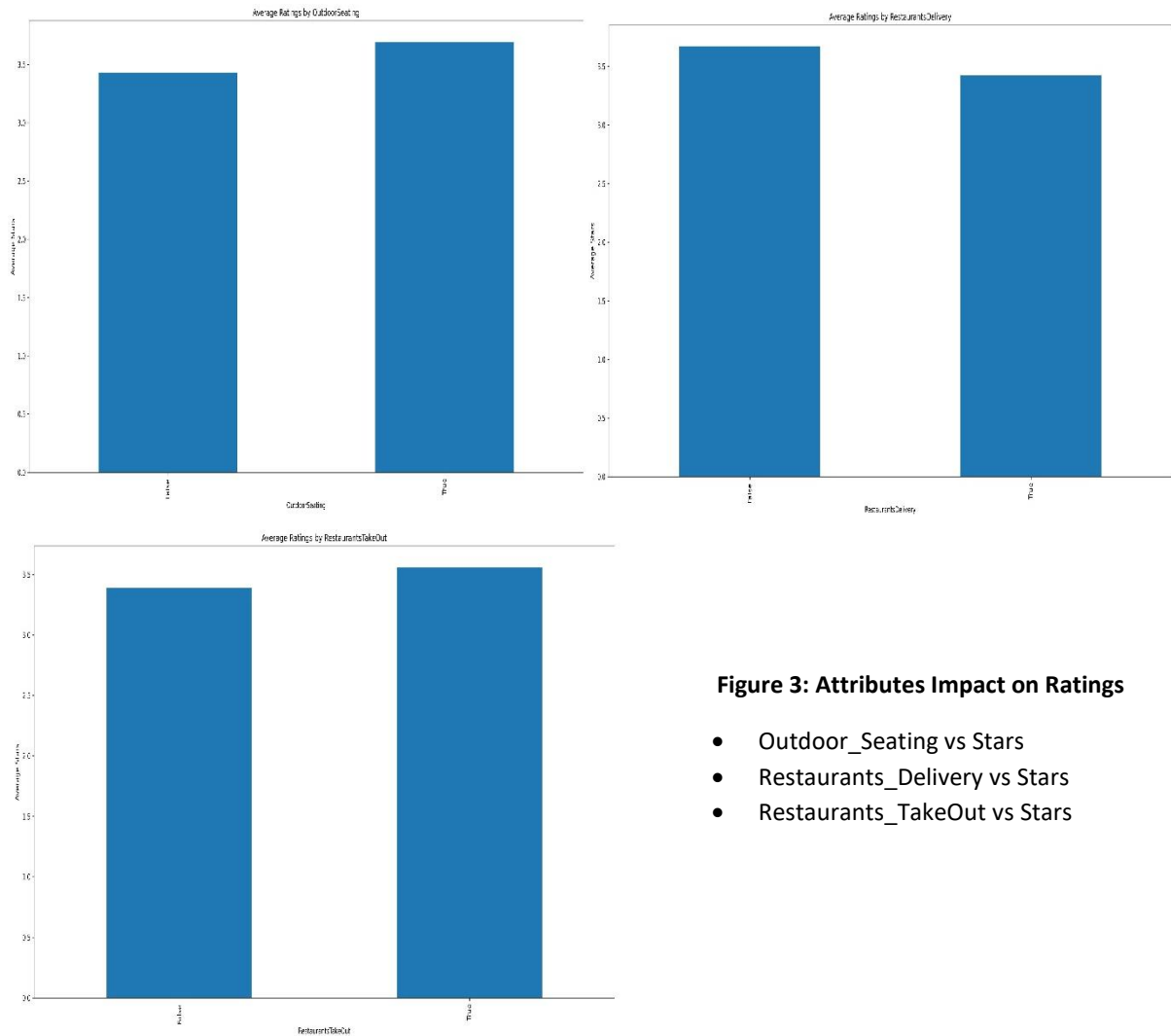


Figure 3: Attributes Impact on Ratings

- Outdoor_Seating vs Stars
- Restaurants_Delivery vs Stars
- Restaurants_TakeOut vs Stars

- Higher review counts are weakly positively correlated with ratings
- Lower-to-mid price range businesses achieve better ratings
- Delivery, takeout, outdoor seating improve ratings
- These relationships were evaluated using correlation analysis and grouped average rating comparisons across price ranges and service attributes.

3.2 Multi-Collinearity

Table 4:
Correlation
Table

Variable	Stars	Review_Count	PriceRange	No_of_Categories
Stars	1.000000	0.060747	0.097265	0.056469
Review_Count	0.060747	1.000000	0.075641	0.103016
PriceRange	0.097265	0.075641	1.000000	0.098457
No_of_Categories	0.056469	0.103016	0.098457	1.000000

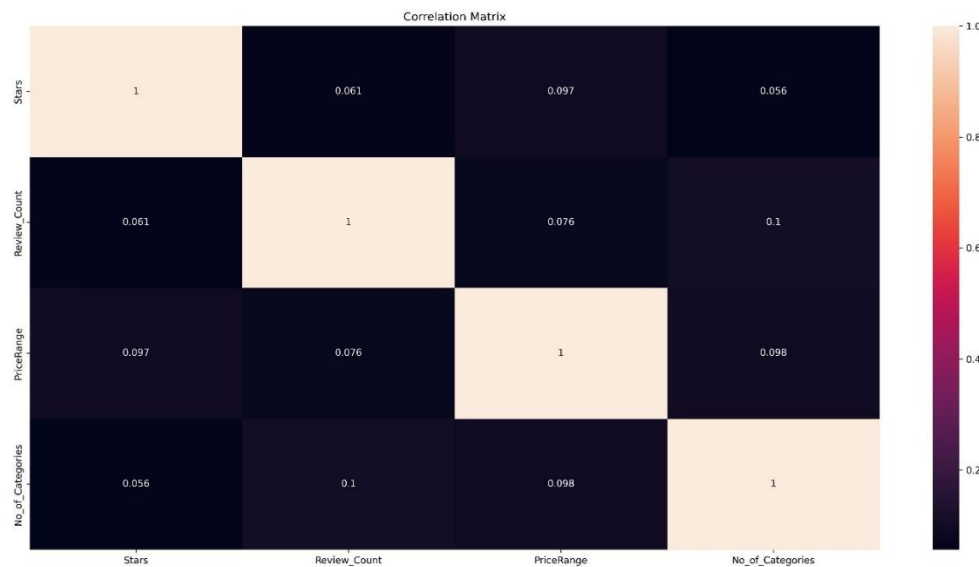


Figure 4: Correlation Matrix

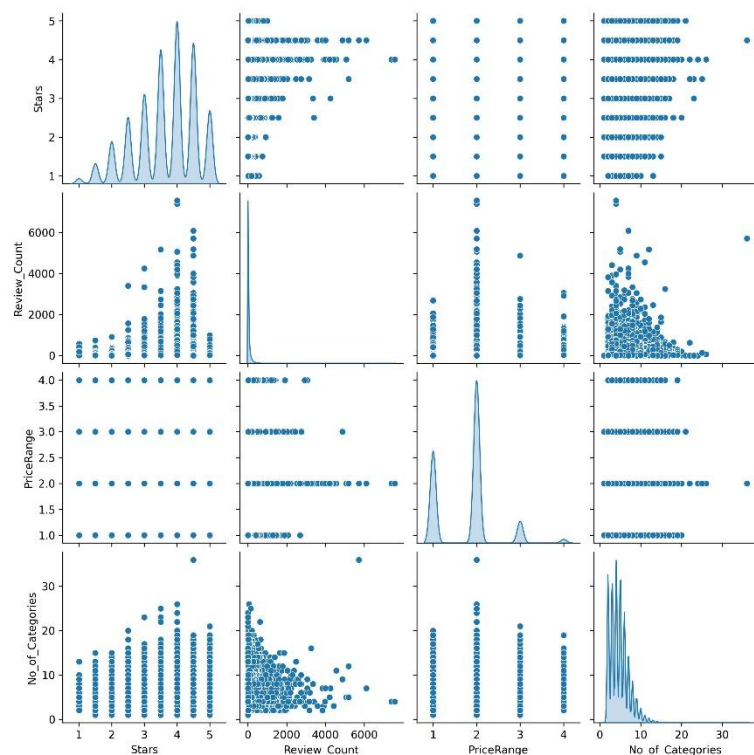
- No severe multi-collinearity observed among numeric variables
- Category bins and attributes show minimal correlation
- Low multicollinearity supports the use of both regression and tree-based models without instability in coefficient estimates.
- This also indicates that dimensionality reduction techniques such as PCA are not mandatory at this stage.

3.3 Pairplot for Key Numerical Variables

Figure 5: Pairplot

(Important Variables)

Pairplot analysis reveals weak correlations among numerical variables, confirms long-tailed review distributions, and indicates no multicollinearity in the dataset.



3.4 Distribution Analysis

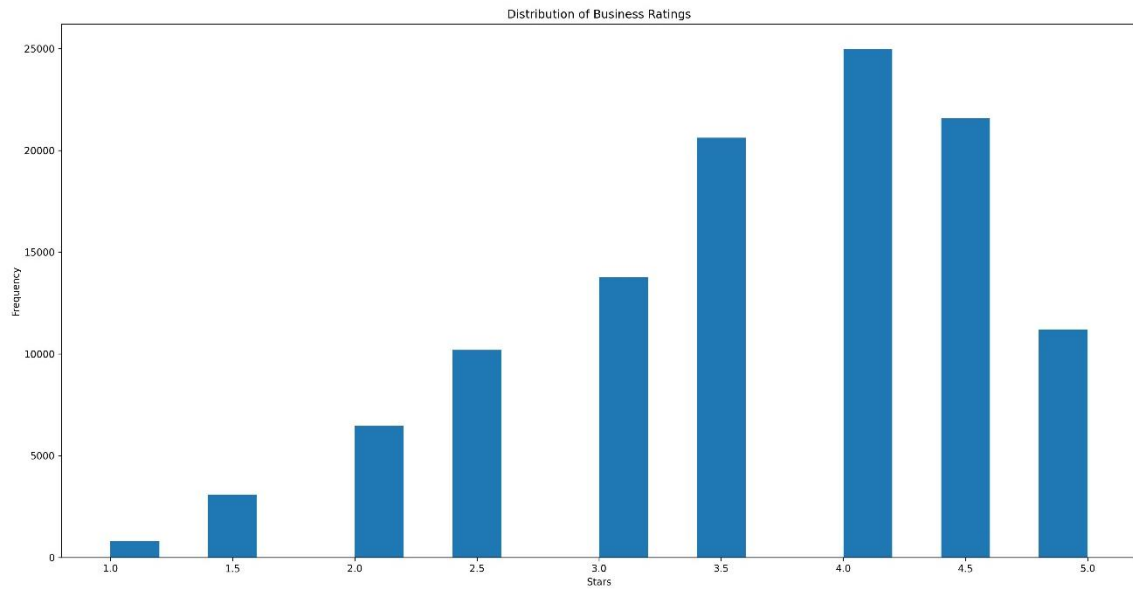


Figure 6: Rating Distribution

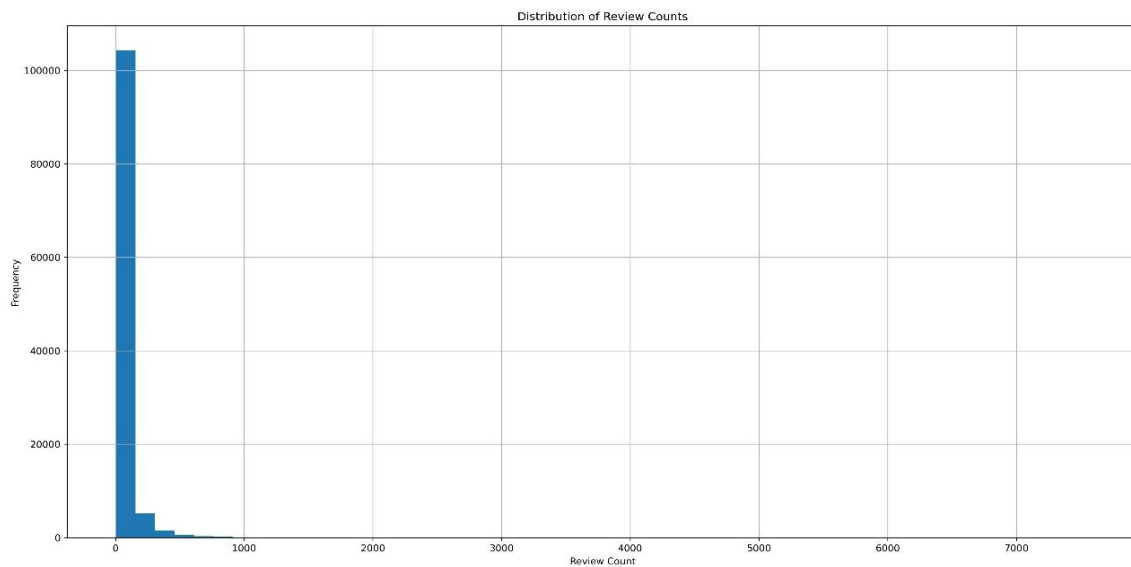


Figure 7: Review Count Distribution

- Ratings are skewed toward higher values (3.5–4.5).
- Review count exhibits long-tail distribution.
- Log transformation was explored but not applied, as tree-based models can effectively handle skewed distributions.

3.5 Outliers

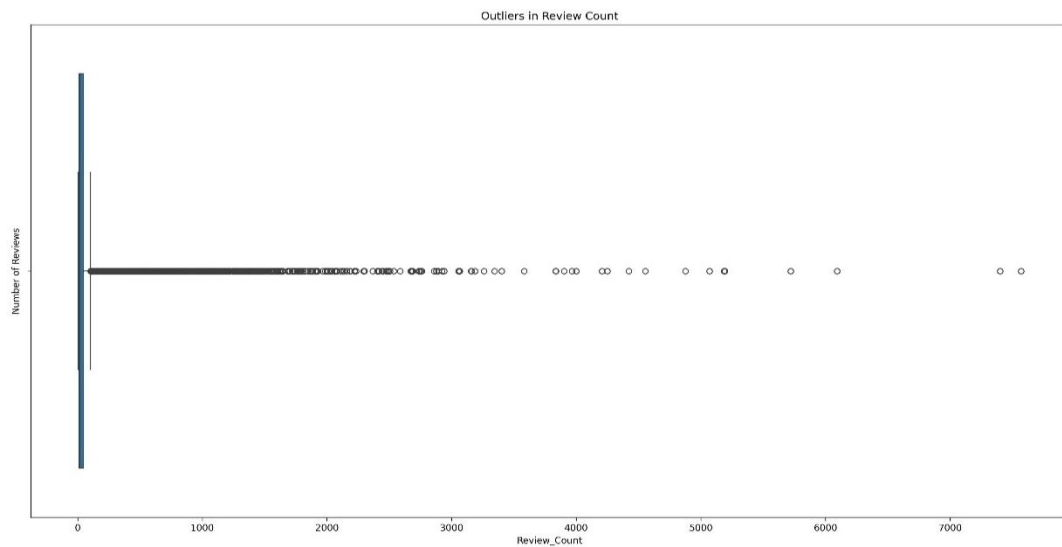


Figure 8: Outliers Detection For Review_Counts

- Present in review counts.
- Treated implicitly using tree-based ML models.
- Extreme values represent genuine high-visibility businesses and were therefore retained.

3.6 Statistical Significance

T-tests and ANOVA confirm attributes like Delivery, WiFi, Outdoor Seating significantly affect ratings.

3.7 Class Imbalance

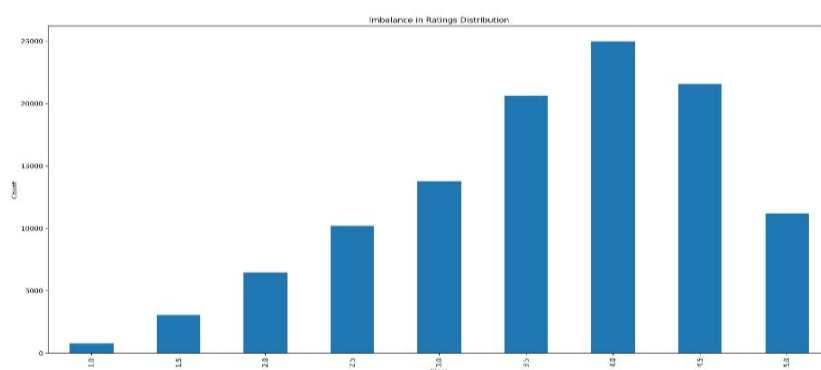


Figure 9: Class Imbalance

- Ratings are imbalanced toward higher scores
- Handled using regression models rather than classification
- Due to the ordinal and imbalanced nature of ratings, the problem is framed as a regression task rather than a multi-class classification problem.

Stars	Count
1.0	789
1.5	3,081
2.0	6,460
2.5	10,193
3.0	13,769
3.5	20,620
4.0	24,967
4.5	21,574
5.0	11,201

Table 5: Class Imbalance

4. Feature Engineering

4.1 Transformations Applied

Step	Description
Boolean Encoding	Converted True/False → 0/1
WiFi Encoding	One-hot encoding
Category Encoding	One-hot category bins
Missing Handling	Median & zero imputation

Table 6: Data Transformation

Several transformations were necessary to convert raw business data into machine-learning-ready features:

- Boolean attributes (e.g., Delivery, TakeOut, OutdoorSeating, GoodForKids) were converted into binary numerical values (0/1).
- Categorical variables such as **WiFi type** and **business category bins** were converted using **one-hot encoding**.
- A derived numerical feature, **No_of_Categories**, was created to capture business diversity.
- Missing values in optional attributes were handled using **median or zero imputation**, while core variables were preserved without alteration.

These transformations ensured compatibility with machine learning algorithms while retaining business meaning.

Feature transformations were selected to balance predictive performance and interpretability. Binary encoding preserved the presence or absence of services, while one-hot encoding ensured categorical variables did not introduce unintended ranking assumptions.

Highly skewed variables, particularly *Review_Count*, were analyzed but not log-transformed. This decision was made because **tree-based models** used in the study are robust to skewness and extreme values, allowing genuine high-visibility businesses to be retained without distortion.

4.2 Scaling

- Scaling was **not applied** during initial modeling, as **tree-based models** (Decision Tree, Random Forest) are invariant to feature scale.
- Scaling will be applied when required for:
 - Linear Regression
 - Support Vector Machines

- Distance-based algorithms such as K-Means clustering

Standard scaling techniques such as **StandardScaler** or **MinMaxScaler** will be used where applicable.

4.3 Feature Selection

- Feature importance scores obtained from a **Random Forest model** were used to evaluate predictor relevance.
- Low-impact and redundant features were removed to reduce dimensionality and noise.
- The selected features improved model interpretability and reduced the risk of overfitting.

This approach combines predictive strength with model explainability.

Feature selection was performed using **Random Forest feature importance** scores. Low-impact and redundant features were removed to reduce noise and improve generalization. This approach enabled the retention of influential predictors while maintaining model interpretability.

4.4 Dimensionality Reduction

- Dimensionality reduction was **not required** during the initial regression modeling phase due to low multicollinearity among numerical variables.
- **Principal Component Analysis (PCA)** is planned in the clustering stage to:
 - Reduce high-dimensional one-hot encoded features
 - Improve cluster separation and visualization
 - Retain maximum variance with fewer components

4.5 Feature Leakage Prevention

To prevent target leakage, rating-related variables were excluded from unsupervised learning and feature selection processes. This ensured that learned patterns remained unbiased and generalizable.

4.6 Model Readiness Validation

The final engineered feature set was validated for compatibility across **linear, tree-based, and distance-based models** by applying transformations, scaling, and dimensionality reduction selectively based on model requirements.

5. Assumptions

5.1 Regression Models

Simple Linear Regression (SLR) & Multiple Linear Regression (MLR)

Assumption	Status	Justification
Linearity	Partially satisfied	Relationships between predictors and ratings are weakly linear but acceptable for baseline modeling
Independence of errors	Assumed	Each business is treated as an independent entity
Normality of residuals	Not strictly satisfied	Real-world ratings are skewed; tree-based models are preferred where strict normality fails
Homoscedasticity	Approximately satisfied	Variance is reasonably stable across fitted values
Multicollinearity	Satisfied	Correlation analysis shows no severe collinearity among numerical variables

Logistic Regression

Assumption	Status	Justification
Independence of observations	Satisfied	Businesses are independent units
Linearity of log-odds	Approximate	Some predictors show non-linear effects
Multicollinearity	Satisfied	Checked using correlation and VIF analysis
Balanced classes	Not satisfied	Rating classes are imbalanced

5.2 Classification Models

Decision Tree

- No assumptions of linearity or normality
- Can handle non-linear relationships and outliers effectively

Random Forest

- No strict distributional assumptions
- Robust to noise, missing values, and outliers

- Reduces overfitting through ensemble averaging

Support Vector Machine (SVM)

- Assumes scaled numerical inputs
- Sensitive to feature magnitude
- Performance affected by class imbalance

Bagged and Boosted Models

- Minimal statistical assumptions
- Performance improves through ensemble learning
- Can handle complex non-linear interactions

5.3 Clustering

Principal Component Analysis (PCA)

Assumption	Status	Justification
Linearly correlated features	Satisfied	PCA applied to reduce redundancy in one-hot encoded features
Feature scaling	Required & applied	Ensures equal contribution of variables
Multicollinearity	Addressed	PCA helps resolve correlated inputs

K-Means Clustering

Assumption	Status	Justification
Numerical input features	Satisfied	Categorical variables encoded numerically
Feature scaling	Required & applied	Distance-based algorithm
Presence of outliers	Partially addressed	Outliers retained as genuine cases; impact minimized via scaling
Spherical clusters	Approximate	Suitable for exploratory segmentation

-----**Thank You!**